Opinion Network Formation

A similarity network built from a class survey on
Technology, Education, Ethics, and Environment

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DPCN, Assignment 1

Team: Deadlines Passing, Care Not

https://github.com/avi1o1/DPCN_A1

Overview

Task: build a network from a class survey on four topics, then analyse it.

Approach: a respondent is a node. An edge joins two people whose survey answers are similar. Three sets of analysis run on that network.

- *Structure.* Degree, clustering, components, four centralities, and Louvain communities.
- *Dynamics.* The DeGroot opinion average and an SIS contagion, both from the course.
- *Null models.* Erdős–Rényi, a degree-preserving rewiring, and a label permutation. These test whether the structure is more than chance.

Headline: 51 of the 87 respondents (59%) form one connected group. The other 36 have no strong-similarity partner at all.

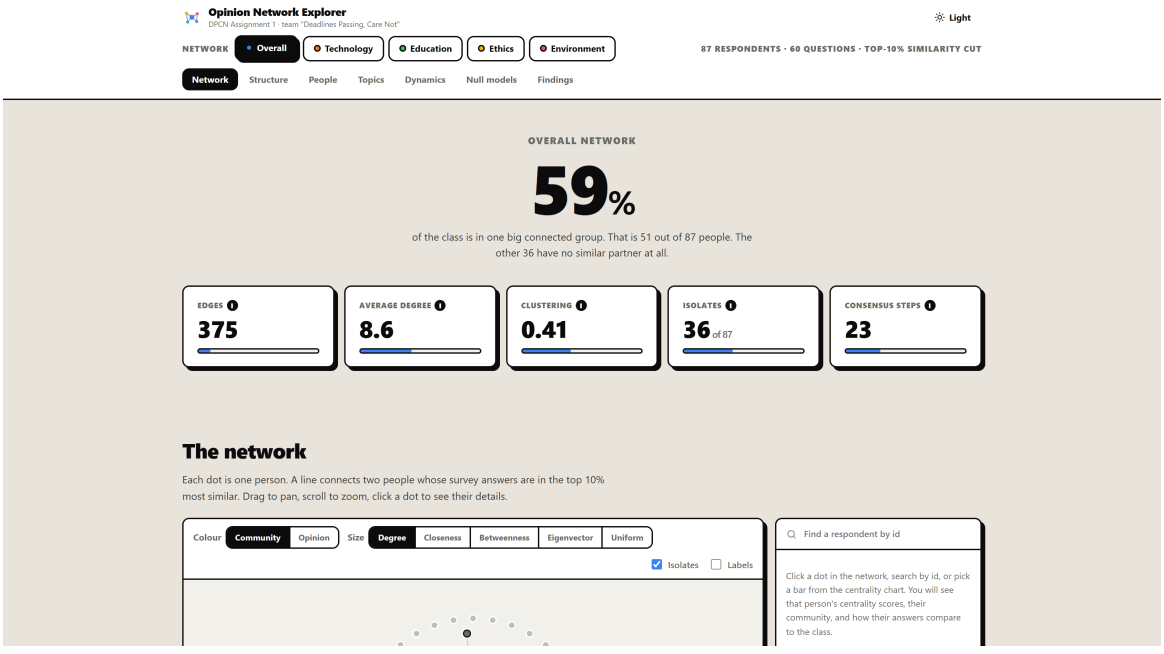


Figure 1. The interactive dashboard. It opens in any browser and needs no server. The network picker changes every number, chart, and diagram to the chosen network.

1. Dataset Documentation

The raw survey: 96 responses to 60 opinion statements, 15 in each of four topics. A letter code marks the topic: **T** Technology, **E** Education, **S** Ethics, **V** Environment. Each statement uses a five-point Likert scale, plus a "No Comments" option.

Preparation:

- *Likert to number.* Strongly Disagree $= -2$ through Strongly Agree $= +2$. The scale is symmetric, so 0 is a true neutral and a mean score carries a direction and a strength at once.
- *"No Comments" is missing, not neutral.* A refusal to answer is not the same as being neutral. There are 62 blanks out of $87 \times 60 = 5220$ (1.19%), spread over 19 respondents.
- *Quality filter.* Drop any respondent who answered fewer than half of the 60 questions. This removes 9 responses and leaves **87 respondents**.
- *Result.* An 87×60 table of scores in $[-2, +2]$, with a blank for each "No Comments". Each question keeps its topic and its text.

Missing values: two rules are correct and both are used on purpose.

- Work that must agree with the graph replaces a blank with 0 before averaging. This is the step the edge builder uses, so opinions and edges come from the same numbers.
- Descriptive statistics drop the blank from a person's average instead.
- The two rules differ by at most 0.01 on the class mean, and by 0.23 for one respondent. A number from one rule is never compared with a number from the other.

2. Pipeline Followed

Similarity: each respondent becomes a vector of 60 answers, $\mathbf{x}_i \in [-2, 2]^{60}$, with blanks set to 0. The similarity of two people is the cosine of the angle between their vectors,

$$s_{ij} = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \, \|\mathbf{x}_j\|} \in [-1, 1].$$

The cosine rewards a shared pattern across all 60 questions. It does not reward a shared average level alone.

Edges: compute the full 87×87 cosine matrix, then keep the top **10%** of pairs as edges. That is the 90th-percentile cut.

- A hard cut is necessary. Almost everyone in this class agrees on average, so a loose threshold gives a near-complete graph with no information.
- The rule keeps a fixed *rank*, not a fixed score. It therefore yields exactly **375 edges** every time, whatever the similarity distribution looks like.
- **Five networks** use the same rule. One *overall* network uses all 60 questions. Four use one topic each, 15 questions apiece.
- Because node count, edge count, and density are identical across the five, any difference between them is a wiring effect. Section 3.5 tests the cut itself.

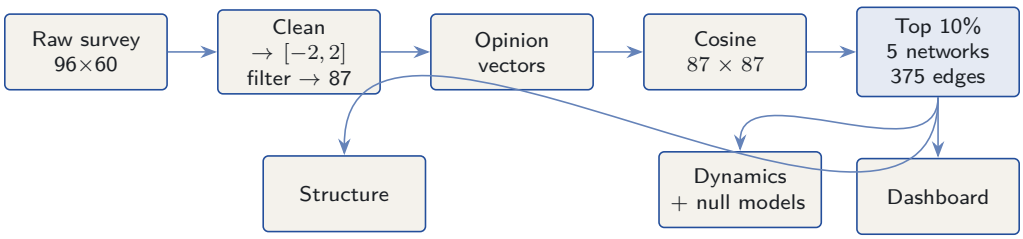


Figure 2. The pipeline. Clean the survey, build the opinion vectors, compute the cosine matrix, cut the top 10% of pairs into five networks. Structure, dynamics, and dashboard all read those saved networks. No later stage rebuilds the graph.

3. Analysis and Visualizations

3.1 Structure

Definitions: *degree* k is the number of edges at a node. *Density* is the fraction of possible edges present. *Clustering* measures how often two neighbours of a node also connect to each other, on a scale of 0 to 1. A *component* is a set of nodes that can all reach each other, and the largest one is the *giant component*. *Modularity* Q scores a grouping by how many more edges fall inside groups than a random graph with the same degrees would give.

Results:

Metric	Overall	Technology	Education	Ethics	Environment
Nodes / Edges	87 / 375	87 / 375	87 / 375	87 / 375	87 / 375
Density	0.100	0.100	0.100	0.100	0.100
Avg. degree $\langle k \rangle$	8.62	8.62	8.62	8.62	8.62
Avg. clustering	0.408	0.387	0.396	0.363	0.375
Components	37	21	23	42	43
Giant component	58.6%	74.7%	72.4%	51.7%	50.6%
Giant diameter	4	5	6	4	5
Giant avg. path	1.88	2.20	2.24	1.78	1.93
Modularity Q	0.108	0.242	0.263	0.151	0.079

Table 1. Structural metrics per network. Node count, edge count, and density are identical by construction, so every other difference is an opinion effect.

Analysis:

- The overall network is sparse but locally dense. Density is 0.10 and clustering is 0.41. Connected people sit in tight triangles.
- **36 of 87 respondents are isolates.** An isolate has no strong-similarity partner anywhere in the class.
- The giant component is compact. Two connected people are 1.88 steps apart on average, with a diameter of 4.
- The degree distribution is heavy-tailed. Respondent 114 has $k = 35$ while the mean is 8.6, a value that fits almost nobody (Figure 3).

3.2 Centrality and communities

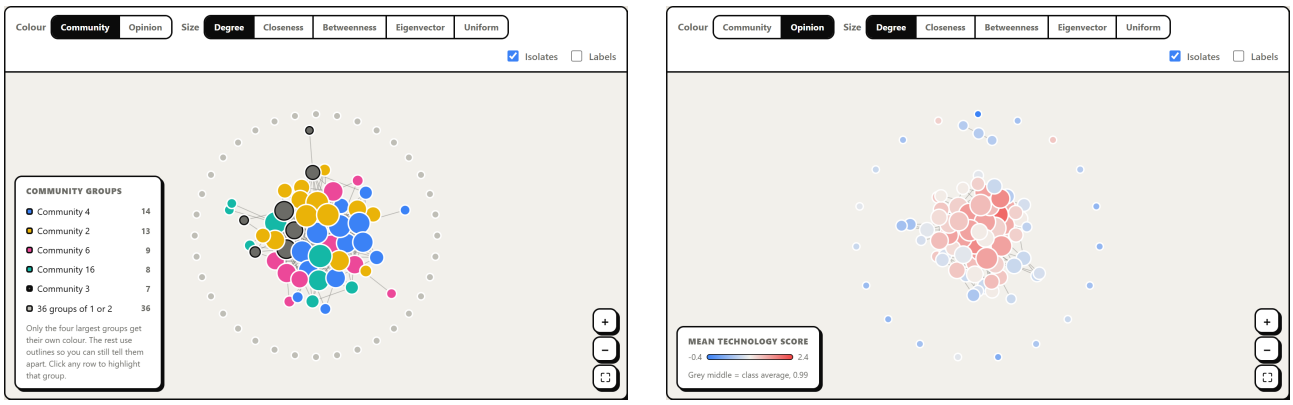
Definitions: *degree* centrality counts direct connections. *Closeness* measures how short the average path is to everyone else. *Betweenness* counts how often a node lies on a shortest path between two others, which rewards a bridge position. *Eigenvector* centrality gives more weight to a node connected to other important nodes. *Louvain* searches for the grouping with the highest modularity.

Analysis:

- Degree, closeness, and eigenvector centrality pick almost the same five people: 114, 67, 94, 105, 110. All five score above the class mean on every topic. They are the uniform strong-agreement answerers, so their 60-value vectors resemble many others.
- Betweenness gives a different answer. Its top node is **41**, who has a maximum Technology score but only an average Education score. That mixed profile bridges sub-groups.
- The most connected person and the best bridge are different people. One centrality measure is not enough to describe importance here.
- Louvain finds many small communities and a low modularity ($Q = 0.108$). Every community has a positive mean on every topic. Only one group (C6) falls below the class mean anywhere, and only on Technology. These are degrees of agreement, not opposing camps.



Figure 3. *Left:* degree distribution on log-log axes as a complementary CDF $P(K \geq k)$. The 36 isolates sit at $k = 0$, which a log axis cannot draw, so the chart states this rather than dropping them silently. *Right:* the ten most central respondents by degree.



(a) Overall, coloured by community. (b) Technology, coloured by opinion.

Figure 4. Node-link view, node size scaled by degree. The giant component sits in the centre, small components ring it, isolates sit on the outer rim. **(a)** The four largest Louvain communities are coloured, singletons and pairs are grey. **(b)** Strong-agreement answerers (red) form the centre; lukewarm answerers (blue) sit on the edge.

3.3 Cohesion tracks opinion variance, not opinion strength

Observation: the per-topic numbers run against the raw averages.

Topic	Class mean	Std. dev.	Giant component	Components
Technology	0.99	0.37	74.7%	21
Education	0.91	0.35	72.4%	23
Ethics	1.28	0.45	51.7%	42
Environment	1.37	0.47	50.6%	43

Table 2. The topics with the strongest average agreement have the most broken networks. The spread explains it, not the mean.

Analysis:

- Ethics (+1.28) and Environment (+1.37) have the strongest average agreement, yet the most broken networks at 52% and 51%.
- Technology (+0.99) and Education (+0.91) agree less on average, yet stay the best connected at 75% and 72%.
- The cause is the spread. Cosine similarity depends on the pattern across the 15 questions of a topic, not on the average level.
- Technology and Education opinions are tightly grouped (s.d. 0.37 and 0.35), so most vectors stay similar and the graph holds together. Ethics and Environment are strong but spread out (s.d. 0.45 and 0.47), so the same cut fragments them.
- "The class agrees more on Environment than on Technology" is true of the raw average and false of the network.

3.4 Dynamics on the network

DeGroot: give each respondent a start opinion, their mean score. At each step every respondent replaces it with the mean of their neighbours' opinions, all at the same time. The rule is $\mathbf{x}(t{+}1) = D^{-1}A\,\mathbf{x}(t)$. The eigenvalues of the update matrix decide the outcome. The course stability rule says a linear map is stable when the relevant eigenvalue has magnitude below 1. The *SLEM* (second-largest eigenvalue modulus) sets the speed, and the algebraic connectivity is $\lambda_2 = 1 - \mu_2$.

Results:

Network	λ_2	SLEM	Steps	Core size	Consensus
Overall	0.627	0.397	23	51	1.409
Ethics	0.455	0.545	31	45	1.350
Technology	0.318	0.682	53	65	1.313
Education	0.264	0.736	61	63	1.275
Environment	0.229	0.771	77	44	1.349

Table 3. DeGroot consensus. Every SLEM is below 1, so every map is stable. The order of λ_2 matches the order of the measured steps with no exception.

Analysis:

- The overall network has 37 components, so there is no single class consensus. The 51-person core reaches **1.409** in **23 steps**.
- The 36 isolates never move. An isolate has no neighbour to average with, so its opinion at step 1000 equals its opinion at step 0.
- Convergence speed follows the algebraic connectivity exactly across all five networks. This is the course stability rule confirmed on survey data.
- Edge weights have almost no effect. The weighted consensus is 1.4099 against an unweighted 1.4085. Row normalisation removes the narrow 0.85 to 0.95 range of the kept similarities.

SIS: model a strong opinion as a contagion. An infected person infects each neighbour at rate β and

recovers at rate γ. The course result gives a critical ratio $(\beta/\gamma)_c = 1/\lambda_{\max}$, where $\lambda_{\max}$ is the largest eigenvalue of the adjacency matrix. A lower threshold means an opinion spreads more easily. Note this uses A itself, not the row-normalised $D^{-1}A$, because an infected person infects each neighbour separately rather than averaging.



Figure 5. *Left:* DeGroot. The connected core funnels onto 1.41 while the isolate lines stay flat. *Right:* the simulated SIS transition begins at about 1.6× the predicted bound, which is expected because the mean-field value is a lower bound and the core holds only 44 to 65 people. *Bottom:* the threshold sensitivity sweep.

3.5 Does the 10% cut-off matter?

Experiment: apply the same rule at every quantile from 0.50 to 0.99 and record the structure at each one.

Results:

Quantile kept	Edges	Isolates	Giant component	Clustering
0.90 (used)	375	36	58.6%	0.408
0.80	749	21	73.6%	0.579
0.70	1123	15	82.8%	0.643
0.60	1497	13	85.1%	0.674
0.50	1871	8	90.8%	0.751

Table 4. Loosening the cut connects most of the isolates but never all of them, and inflates the clustering until the graph stops being informative.

Analysis:

- The topic order is stable at every cut. Technology stays above Environment and Education stays above Ethics, with no exception. That order is a property of the data, not of the 90th percentile.
- The isolates are only partly an artifact. Their count falls from 36 at $q = 0.90$ to 8 at $q = 0.50$, and never reaches zero. Eight people stay below the median similarity to every other person in the class.
- A looser cut has a cost. At $q = 0.50$ the clustering rises to 0.75 and the graph is half complete, so community detection and centrality stop meaning much.
- $q = 0.90$ is therefore a tested compromise rather than an assumption.

4. Results and Discussion

What the network says about class opinions: this class is agreeable. The mean opinion is positive on all four topics and the raw data shows no opposing camps. The network exposes structure that the averages hide.

Finding 1. No global consensus, and the drop-outs are the moderates.

- Under DeGroot the 51-person core agrees at 1.409 while the 36 isolates never move.
- The core started at a mean of 1.309. The isolates started at 0.872.
- So the process removes the more moderate half of the class. This is a structural effect of the construction, not something the averaging model causes.

Finding 2. Cohesion is a variance story. Technology and Education look better connected than Ethics and Environment despite lower mean scores, because their opinions are less spread out. Raw agreement and social cohesion are not the same measurement.

Finding 3. Spread is easiest where the class is most broken.

Network	$\lambda_{\max}$	$1/\lambda_{\max}$	Giant size	Giant density
Environment	23.92	0.0418	44	0.395
Ethics	22.45	0.0445	45	0.378
Overall	22.06	0.0453	51	0.294
Technology	18.17	0.0550	65	0.179
Education	17.74	0.0564	63	0.191

Table 5. Epidemic thresholds. The order reverses the connectivity order in Table 1. The most broken topic spreads most easily.

- Environment and Ethics have the lowest thresholds at 0.0418 and 0.0445, yet they are the least connected topics.
- The edge count is fixed at 375, so a smaller giant component packs almost the same edges into fewer people. Environment puts 374 edges into 44 respondents; Technology spreads 373 across 65.
- A denser core raises $\lambda_{\max}$, which lowers the threshold. The standard bound $\langle k \rangle \leq \lambda_{\max} \leq k_{\max}$ applied inside the giant component predicts this ordering directly.
- Fragmentation and easy spread are not opposites here. This is a real property of the data and also a warning about the fixed-edge method.

Finding 4. The structure is in the degree sequence. A null model asks whether a random network with no opinion structure gives the same value. The z-score is $z = (\text{real} - \text{null mean})/\text{null s.d.}$, and $|z| < 2$ means the real value looks random.

- Against 500 ER graphs the network is unusual on every measure, with a clustering z-score of $+27$ and a giant-fraction z-score of -362.
- Against 500 degree-preserving rewirings it is ordinary. Clustering, giant component, and modularity all fall inside $|z| < 2$.
- So once each person's edge count is fixed, chance supplies the rest of the structure.
- For this construction that is the same statement as "the connected core is the group that agrees with everything". Agreement level sets the degree, and after the degrees are fixed nothing is left to explain. This gives the reason for the structural analysis rather than weakening it.

Statistic	Real	ER mean	z	Rewired mean	z
Average clustering	0.408	0.098	+27.2	0.391	+1.6
Giant fraction	0.586	1.000	−361.7	0.586	0.0
Average path length	1.880	2.292	−9.5	1.834	+3.7
Modularity	0.105	0.282	−13.7	0.100	+0.9

Table 6. The real network against 500 Erdős–Rényi graphs of the same density and 500 degree-preserving rewirings of itself. Green marks a z-score inside the random band.

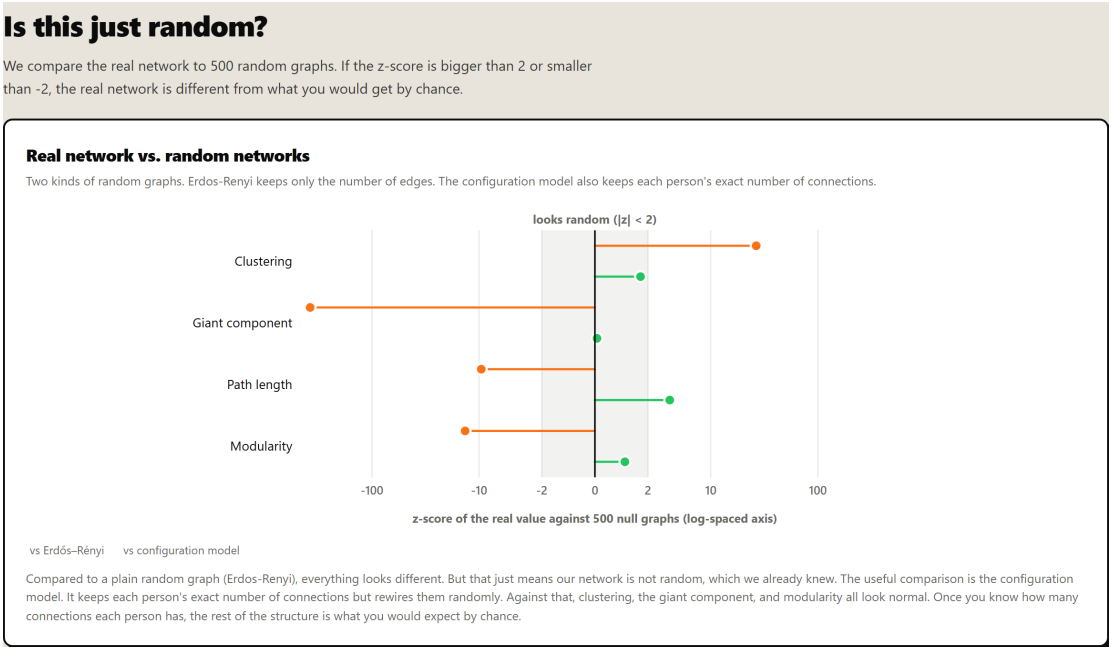


Figure 6. Null-model z-scores on a symmetric-log axis. The shaded band is the $|z| < 2$ range that looks random. Erdős–Rényi fails every measure. The configuration model matches clustering, giant component, and modularity.

Finding 5. The communities are real, but only at the fine grain. A permutation test shuffles the community labels 10,000 times with the group sizes held fixed, then measures the opinion variance inside the groups. The p-value is the fraction of shuffles that beat the real grouping.

Features tested	Observed	Null mean	z	p
All 60 question scores	0.4116	0.4592	−6.57	< 0.0001
4 topic means	0.0791	0.0839	−1.26	0.108

Table 7. The same communities pass on the answers the graph was built from and fail on the coarser topic averages.

- On the 60 question scores the Louvain grouping is far tighter than chance ($z = -6.57$, $p < 0.0001$).
- On the 4 topic means it is not distinguishable from random ($z = -1.26$, $p = 0.108$).
- Louvain groups people by their detailed answer pattern, not by their broad position on the four topics. Two people in one community can differ on topic averages while answering the individual questions alike.
- The community-by-topic heatmap therefore *describes* each group but does not *define* it, and is not presented as a definition.

In summary:

- The class is agreeable across all four topics, with no real factions.
- Low variance drives the apparent cohesion, so Technology and Education out-connect Ethics and Environment even with lower means.
- The connected core is a behaviour type, the uniform strong-agreement answerers, not a demographic

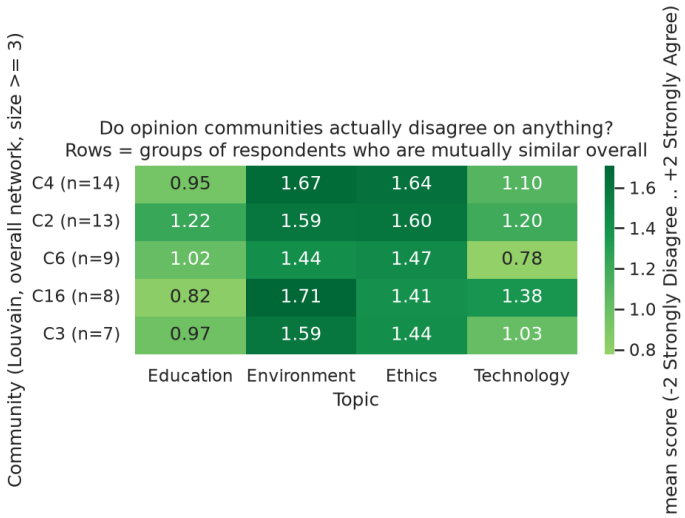


Figure 7. Mean score of each large community, per topic. Every cell is positive. The communities raise or lower agreement rather than disagreeing.

or a topic camp.

– Centrality measures disagree about who matters, and that disagreement is itself a result.
– Ethics and Environment move together at a respondent-level $r = 0.70$ and form one values axis. Technology and Education are nearly independent at $r = 0.29$.

5. Individual Contribution

Member	Tasks completed
Aviral	Cleaned and interpreted the survey data. Built all five networks with cosine similarity and the top-10% cut. Ran the structural analysis: degree, clustering, components, the four centralities, and the Louvain communities. Produced the static figures. Found the topic-variance result and the centrality result.
Varun	Ran the dynamics and the spectral analysis. Ran the DeGroot average with the Laplacian stability analysis. Computed the SIS epidemic threshold and its simulation. Built the Erdős–Rényi and configuration null models. Ran the community permutation test and the edge-threshold sensitivity sweep.
Arihant	Built the self-contained interactive dashboard: hand-drawn charts, a node-link view, all five networks, community and opinion colour, an isolate toggle, a respondent lookup, and the dynamics and null-model panels. Exported the dashboard data. Wrote this report and produced its figures.

Table 8. Individual contributions. The commit history matches this split.

Reproducibility: everything in this report comes from the linked repository. The pipeline runs as a fixed sequence of stages and writes its results to disk, so no figure or table is computed by hand. The dashboard is a single HTML file that opens in any browser with no server, no build step, and no network access.